Target-Aspect Domain Continual Learning for SAR Target Recognition

Hongting Chen, Chuan Du, Jinlin Zhu, and Dandan Guo

***Abstract*— In recent years, impressive progress has been achieved in synthetic aperture radar (SAR)-based automatic target recognition (ATR) with the development of deep learning. In practice, a complete training SAR image dataset in all target-aspect domains is limited available for one measurement. When SAR images in the unseen aspect domains are newly acquired, direct retraining of the trained SAR-ATR models with them may lead to a significant performance decline for the seen aspect domains. In this article, we propose an aspect continual recognition model (ACRM) to address the learned feature forgetting problem when SAR images with different target aspects come sequentially in the real-world SAR-ATR. Considering the abundant variations of SAR images with target aspects, we introduce the Bayesian probabilistic frame to improve the model's generalization of characterizing the varied target features across different aspects. To acquire a better solution for the posterior probability of the model parameters, we integrate an online Monte Carlo variational inference into the deep neural network in the ACRM. Furthermore, to mitigate the accumulation of estimation errors caused by the repetitive approximations in inference, we leverage the coreset method by retaining a small subset of important samples from previous tasks as a coreset. We conduct extensive experiments on the MSTAR and FUSARship datasets. Compared with a variety of baseline algorithms in continual learning, our methods exhibit excellent SAR-ATR performance and robustness, when the SAR images from different target aspects are acquired sequentially.**

***Index Terms*— Automatic target recognition (ATR), online learning, synthetic aperture radar (SAR), target-aspect continual learning.**

I. Introduction

DUE to its all-weather, all-day imaging capability, synthetic aperture radar (SAR) has become a valuable tool in Earth surveillance. Therefore, SAR images are widely used for target recognition across military and civilian applications. The development of SAR imaging technology has led to the extensive use of airborne and spaceborne SAR systems, resulting in the generation of massive large-scale SAR images, therefore precise SAR-ATR is essential in SAR image interpreting.

Originally, SAR-ATR algorithms mostly rely on template matching [1], [2] or modeling of target electromagnetic characteristics [3], both of which demand extensive expertise. However, in complex scenarios such as heterogeneous background clutters existing in SAR images, it is difficult to get the exact matches for the templates, which leads to a decrease in recognition accuracy. With the development of machine learning, various classification algorithms, such as support vector machines (SVMs) [4], adaptive boosting [5], metric factorizing [6], and dictionary analysis [7], emerge to be utilized for SAR image classification. Confronting complex sea clutter, Chen et al. [8] proposed an SAR ship detection method that mitigates speckle interference, thus enhancing the significance of the target area and recognition performance. However, the actual targets in SAR images have complex scattering center compositions and the above models typically extract only artificial or shallow features. Therefore, the data description capability of these models for SAR-ATR is inadequate.

Deep learning models accomplish robust hierarchical feature extraction via nonlinear mappings across different layers [9], thus endowing them with powerful capabilities in target recognition. Ding et al. [10] utilized the electromagnetic properties of SAR images to construct several data argumentation methods for SAR-ATR based on convolutional neural networks (CNNs), which improve the models' generalization ability and recognition performance. Gao et al. [11] further improved CNN's feature extraction ability by introducing an SVM-CNN hybrid model, which incorporates interclass separability in the loss function and enhances its SAR-ATR performance. Zhanget al. [12] developed a multiscale global scattering feature association module. It fuses scattering and deep network features to produce a more comprehensive SAR ship target representation, which performs well in few-shot learning.

However, when the trained SAR-ATR model encounters target samples from a new category outside the distribution of the training dataset, the performance of the recognition model will degrade significantly, presenting a typical open set recognition (OSR) challenge. Zhao et al. [13] addressed this by mapping features to a spherical space to enable tighter clustering of intraclass features, while Jiang et al. [14] synthesized features from seen classes to simulate unseen classes, ultimately training a $K+1$ classifier for OSR in SAR-ATR. Wang et al. [15] used meta-learning to construct a dynamic feature space that distinguishes between seen and unseen classes and introduced

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entropy-aware loss to enhance the feature space and improve the model's robustness. Furthermore, Li et al. [16] approached the OSR problem by adding fully connected layers to the discriminator of a generative adversarial network, effectively decoupling the tasks of classification and anomaly detection, thereby addressing the challenge of recognizing unseen SAR target categories.

When encountering unseen categories of target samples during the application of the trained SAR-ATR model, subsequent training on new SAR images leads the model to forget the learned knowledge pertaining to the old categories of targets. Therefore, several studies investigate utilizing continual learning to solve the problem of adding new categories of targets, so-called class incremental learning (CIL), ensuring that the model maintains high recognition performance for both new and old categories of targets [17], [18], [19], [20]. Tang et al. [17] proposed the high plastic error correction incremental learning, which adaptively adds new anchored class centers to tackle the challenges of model degradation and declining plasticity in the SAR-ATR CIL task. Gao et al. [18] presented a CIL method, which employs both intra- and interclass samples' scatters to compute the feature separability loss and address the learned knowledge forgetting problem in SAR-ATR. Pan et al. [19] proposed a CIL method in a small set of training samples based on a vision transformer structure, which can produce a self-sustainment guidance representation of SAR image. Li et al. [20] anchored the SAR class centers and made the sample features gather near their corresponding anchor point through the clustering algorithm. In this way, the intraclass distances are minimized and the interclass distances are maximized, thereby the recognition performance and robustness of the SAR-ATR model are improved in CIL.

It is evident that a complete training SAR image dataset covering all target-aspect domains is not available for a single measurement [30]. The above studies are conducted in the framework of a complete target-aspect training dataset. To deal with the problem of target-aspect incremental learning, the SAR-ATR model should possess the ability to learn the knowledge of the targets with new aspects and maintain the learned knowledge of the targets with the previous aspects. When SAR images with different target aspects acquired in the form of data stream, traditional deep learning frameworks often encounter the challenge of catastrophic forgetting, while CIL also faces stability and adaptability dilemmas [17]. The varied target aspects resulting in the significant variations of scattering point distribution in SAR images impose a stringent challenge on the model's generalization ability. In this article, we propose an aspect continual recognition model (ACRM) approach to address the target-aspect feature forgetting problem, when SAR images with different target aspects are acquired sequentially in the real-world SAR-ATR. Considering the varied scattering point distributions in SAR images with different target aspects, we use a Bayesian probabilistic framework to enhance the model's generalization for characterizing different target features across target aspects. To acquire a better solution for the posterior probability of the model parameters, we integrate an online Monte Carlo variational inference into the deep neural network in ACRM. Furthermore,

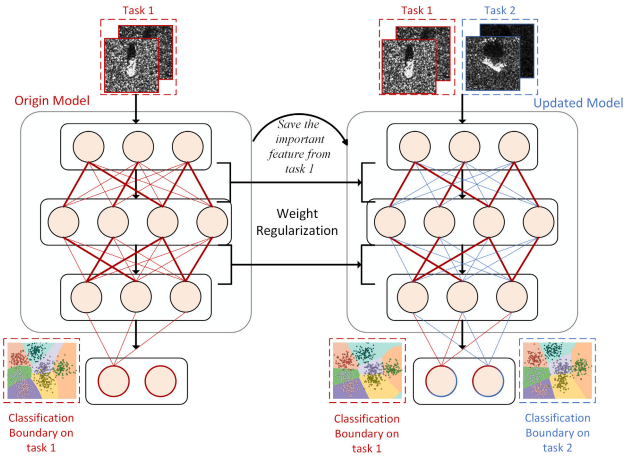


Fig. 1. Basic idea of ACRM in SAR image target-aspect continual learning. The update of the model through the constraints of the parameters is demonstrated by adding explicit regularization terms to mimic the parameters using weight regularization of the old model.

to mitigate the accumulation of estimation errors caused by the repetitive approximations in inference, we leverage the method [21] of retaining a limited number of typical datasets from previous tasks as a coreset. Fig. 1 shows the basic idea of ACRM to achieve online SAR image recognition across different target-aspect domains.

In summary, our contributions are as follows.

1) When the targets' SAR images are acquired sequentially in the domain of the target aspect, the newly collected SAR images will make the trained SAR-ATR model forget partial information learned from the previous target-aspect data and get a decline in recognition performance. To our knowledge, we are the first that propose an SAR-ATR model based on continual learning to solve the important practical problem mentioned above.
2) We utilize a Bayesian framework to model the model parameters. Thus, the model can sufficiently describe the complex and varied scattering point distributions in SAR images with different target aspects, which gives the model a stronger generalization ability. To achieve real-time recognition of SAR images with different target aspects acquired in data streams, we further train the model via online variational inference.
3) A strategy of selecting informative samples from the previous tasks' SAR images to form a coreset is adopted, allowing for a tradeoff between the model's recognition accuracy and computational cost via the changing of coreset size. Furthermore, this strategy can mitigate the cumulative errors resulting from the iterative approximation between model parameters' approximate posterior and their true posterior during the online variational inference.

The rest of this article is organized as follows. Section II discusses several studies related to SAR target classification intending to enhance model generalization, which includes transfer learning, meta-learning and continual learning. Section III provides detailed descriptions of each component in the ACRM. Experimental results and analyses

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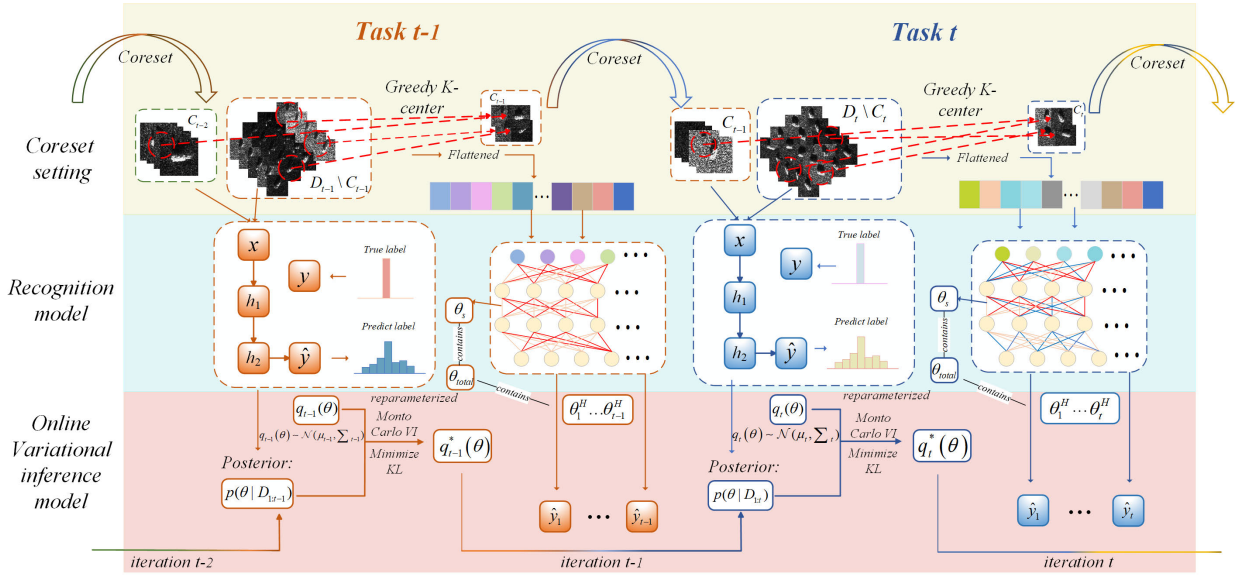


Fig. 2. Structure of overall ACRM.

are presented in Section IV. Section V offers a comprehensive summary of this article.

II. Related Work

SAR images are widely utilized for real-time monitoring, requiring accurate target identification when SAR images with different target aspects arrive in the form of data streams. Consequently, the generalization capability of the recognition model is of paramount importance. There are many ways to improve the model's generalization ability in the presence of lacking training data, such as transfer learning [22], [23], [24], meta-learning [26], [27], [28], [29], or continual learning [35], [36], [37], [38], [40].

A. Transfer Learning

Transfer learning facilitates the use of knowledge from source tasks to enhance predictions in a target task [22]. In SAR-ATR, when training SAR images are scarce, transfer learning often enhances the model's generalization by pretraining it on datasets with related structures or features.

For instance, Malmgren-Hansen et al. [23] explored the transfer learning between a simulated dataset and a set of real SAR images based on the CNN. In this way, targets in the simulated dataset do not have to be geometric replicas of the real SAR image targets, as long as they share similar image features. Huang et al. [24] proposed a domain adaptation method based on multikernel maximum mean discrepancy, which integrates both unsupervised and supervised learning to extract target features from both source- and target-domain data, regardless of label information. This approach enhances the generalization of features in task-specific layers and improves the transferability of features from vehicle data to ship data in SAR-ATR. In the case of limited labeled SAR images and sufficient unlabeled ones, several researchers have been devoted to extracting target features from simulated or optical images and applying them to SAR-ATR task, thus the lack of labeled training SAR images data can be alleviated [23], [24].

B. Meta-Learning

With the knowledge (e.g., average gradient) learned from a large number of training tasks, the model can more quickly adjust its parameters and strategies to the new task's specific requirements with a few samples, thus improving the model's ability to generalize and adapt to new task [25].

Aiming to solve the few-shot problem of SAR-ATR, Fu et al. [26] proposed a meta-learning framework called MSAR, consisting of a meta-learner and base learner. It utilizes appropriate update strategies and learns a good initialization, which can achieve fast adaptation with a few training SAR images on new tasks. To tackle domain shift and task transfer challenges, Wang et al. [27] proposed a model which integrates meta-learning and adversarial domain adaptation to rapidly adjust to new target recognition tasks using real SAR images with refinement. Above all, they leverage a substantial number of simulated SAR images along with a small set of real ones for pretraining the model. Several works combine meta-learning with transfer learning [28], [29] to achieve improved recognition effectiveness with only a few training SAR images in the new task.

Both meta-learning and transfer learning aim to improve the model's performance on new tasks by utilizing the common features in the old and new tasks' SAR images. However, meta-learning focuses on the rapid adaptability from the SAR images of multiple old tasks to the few-shot samples of new tasks, while transfer learning emphasizes transferring knowledge from source task samples to target task samples with similar features [25].

C. Continual Learning

Continual learning facilitates incremental knowledge acquisition from data streams across different domains or tasks, enabling SAR-ATR models to recognize new classes without losing previously learned features [25]. In SAR-ATR,

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most continual learning approaches address incremental class recognition via regularization or replay strategies.

The replay-based methods store a few elaborately selected training samples from the old tasks, which are combined with the new task data samples to constitute the new task training dataset. In the SAR-ATR field, Dang et al. [30] proposed an incremental learning method based on local geometric and statistical information for class boundary example selection, which introduces data reconstruction techniques to update the example set and ensures the completeness of the previous class boundaries when new SAR target classes are added. Oveis et al. [31] utilized the local interpretable model-agnostic explanations algorithm to generate local explanations of SAR images and evaluate the degree of overlap between target and background regions, thereby quantifying the Model's attention to image regions. This method optimizes sample selection and makes the replay samples more representative and meaningful in CIL.

Regularization is used to limit the degrees of freedom of a model by constraining its parameters or output predictions, thereby alleviating the forgetting of previously learned knowledge from previous tasks. Huang et al. [32] compressed the SAR target feature by introducing an attention mechanism and combined cosine similarity to balance the weight deviation of new and old classes. Meanwhile, a hybrid loss function based on knowledge distillation and angle penalty is used to enhance the intraclass compactness and interclass diversity. By combining Euclidean distance and cosine distance, Wang et al. [33] enhanced the diversity between SAR target classes in the embedding space and effectively reduces the intraclass overlap. Meantime, the graph attention network is used to propagate the context information and adjust the classifier weight of each task, to effectively prevent catastrophic forgetting.

Transfer learning and meta-learning can make the model adapt to new tasks, but they cannot prevent the trained model from forgetting the learned target features on those previous tasks. While continual learning learns the target features in the new task and reduces forgetting the learned features from the old tasks [34].

III. Methodologies

Considering the limited availability of a complete training SAR image dataset in all target aspects for one measurement, to online learn and recognize SAR images with new target aspects acquired sequentially, we propose the ACRM. The structure of ACRM is shown in Fig. 2, which is composed of three modules: the coreset module, the recognition model, and the variational inference model. Monte Carlo sampling is used in the variational inference to model the varied scattering point distributions of targets with different aspects in real SAR images. A coreset is embedded in the recognition model to mitigate the accumulation of estimation errors caused by the repetitive approximations in the iterative variational inference.

A. SAR Auto-Target Recognition

For SAT-ATR, SAR image dataset $\{\boldsymbol{x}^{(n)}, y^{(n)}\}_{n=1}^{N}$ is given, where $\boldsymbol{x}^{(n)}$ refers to the nth SAR image, while $y^{(n)}$ represents the nth true label. Our recognition model returns a predicted probability distribution $p(y|\boldsymbol{\theta}, \boldsymbol{x})$ over the target categories with the given SAR images $\boldsymbol{x}$ and the recognition model's parameters $\boldsymbol{\theta}$.

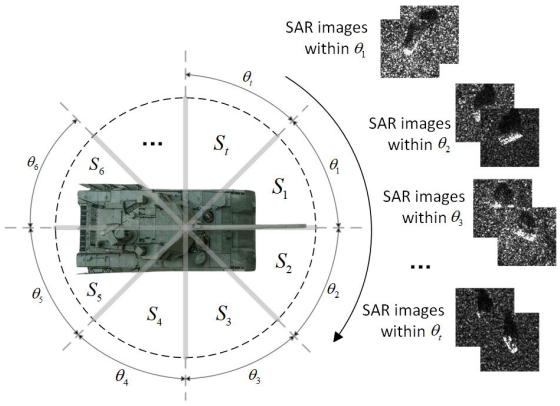


Fig. 3. Segmentation of the SAR image dataset by the target-aspect domain.

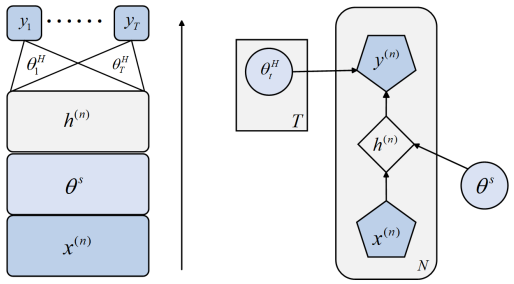


Fig. 4. Schematics of ACRM, including (a) network architecture and (b) graphical model of ACRM. (a) How the multihead recognition model's parameters are shared during training. The lower-level network's parameters $\boldsymbol{\theta}^S$ are shared across multiple tasks. Each task t has its own "head network" parameterized by $\boldsymbol{\theta}_t^H$ mapping from a common hidden layer $h^{(n)}$ to the output y_t. The parameters of ACRM are $\boldsymbol{\theta} = \{\boldsymbol{\theta}_{1:T}^H, \boldsymbol{\theta}^S\}$.

We design a deep CNN classifier and use the cross-entropy loss. The output layer is designed as a softmax layer to derive the predicted target classification probability. The network architecture used in the method can be changed according to the actual application requirements.

B. SAR Continual Recognition Model

1) Recognition Model: In practice, it is difficult to get all target-aspect SAR images at one measurement. We assume the observed SAR images arrive sequentially as a data stream. Since scattering characteristics of SAR targets vary with different target aspects, we define classifying SAR images in a specific target-aspect domain as a distinct task. The SAR images are divided into T target-aspect domains as shown in Fig. 3. So, there are a total of T tasks in our continual learning SAR-ATR. The tth task contains the target-aspect domains of $\{[(360\times(t-1)/T)]^\circ, [(360\times t/T)]^\circ\}, t = 1, \ldots, T$. The SAR image dataset $\mathcal{D}_{1:T}$ across T tasks can be described as

$$\mathcal{D}_{1:T} = \left\{ \left\{ \boldsymbol{x}_t^{(n)}, y_t^{(n)} \right\}_{n=1}^{Nt} \right\}_{t=1}^{T}. \tag{1}$$

In the tth task, there are N_t training SAR images x_t and the corresponding labels y_t.

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The basic classification model is consistent with that in Section III-A, but a multihead network is designed at the output layer of the model to better adapt to the task of target-aspect continual learning. Fig. 4 shows the network architecture and the graphical model of ACRM. ACRMs for different tasks share the lower-level networks across different tasks, which map from observations $\boldsymbol{x}$ to the intermediate-level representations h. However, each ACRM also possesses its own dedicated "head network," connecting the intermediate representations h to the outputs y.

2) Online Variational Inference Model: When SAR images with different target aspects are acquired sequentially, our objective is to online learn the parameters of the recognition model for different target-aspect domains. As described in the above subsection, for Task t, our recognition model returns a predicted probability distribution $p(y_t|\boldsymbol{\theta}, \boldsymbol{x}_t)$ over the target categories with given SAR images $\boldsymbol{x}_t$ and the recognition model's parameters $\boldsymbol{\theta}$. Supposing that the SAR images of each task are statistically independent, given all observed data $\mathcal{D}_{1:T}$, the posterior distribution for model parameter $\boldsymbol{\theta}$ can be expressed as

$$p(\boldsymbol{\theta}|\mathcal{D}_{1:T}) = \prod_{t=1}^{T}\prod_{n=1}^{N_t} \frac{p_0(\boldsymbol{\theta})\, p\left(y_t^{(n)}|\boldsymbol{\theta}, \boldsymbol{x}_t^{(n)}\right)}{p\left(y_t^{(n)}|\boldsymbol{x}_t^{(n)}\right)} \propto p_0(\boldsymbol{\theta}) \prod_{t=1}^{T}\prod_{n=1}^{N_t} p\left(y_t^{(n)} \mid \boldsymbol{\theta}, \boldsymbol{x}_t^{(n)}\right) \tag{2}$$

where $p_0(\boldsymbol{\theta})$ is the prior distribution. $p(y_t^{(n)}|\boldsymbol{\theta}, \boldsymbol{x}_t^{(n)})$ denotes the likelihood distribution of $\boldsymbol{\theta}$.

Note that the above form of recursive posterior obtained by the Bayesian formula is well-suited for continual SAR target recognition, allowing the iterative update of the model's parameters $\boldsymbol{\theta}$, as the new target-aspect SAR images arriving sequentially. According to (2), the recursive posterior can be derived as following:

$$\begin{aligned} p(\boldsymbol{\theta}|\mathcal{D}_{1:T}) &\propto p_0(\boldsymbol{\theta}) \prod_{t=1}^{T} p(\mathcal{D}_t \mid \boldsymbol{\theta}) \\ &= p_0(\boldsymbol{\theta}) p(\mathcal{D}_1 \mid \boldsymbol{\theta}) \prod_{t=2}^{T-1} p(\mathcal{D}_t \mid \boldsymbol{\theta}) p(D_T \mid \boldsymbol{\theta}) \\ &\propto p(\boldsymbol{\theta}|D_1) p(\mathcal{D}_2 \mid \boldsymbol{\theta}) \prod_{t=3}^{T-1} p(\mathcal{D}_t \mid \boldsymbol{\theta}) p(\mathcal{D}_T \mid \boldsymbol{\theta}) \\ &\cdots \propto p(\boldsymbol{\theta}|\mathcal{D}_{1:T-1}) p(\mathcal{D}_T \mid \boldsymbol{\theta}). \end{aligned} \tag{3}$$

Due to the complex form of the posterior distribution $p(\boldsymbol{\theta}|\mathcal{D}_{1:T})$, it is infeasible to evaluate an analytical solution. Inspired by the variational inference theory [43], we build an approximate posterior $q(\boldsymbol{\theta})$ to approximate the true posterior $p(\boldsymbol{\theta}|\mathcal{D}_{1:t})$ as

$$q_t(\boldsymbol{\theta}) \to p(\boldsymbol{\theta}|\mathcal{D}_{1:t}). \tag{4}$$

According to (3), $p(\boldsymbol{\theta}|\mathcal{D}_{1:t})$ is conditioned on $p(\boldsymbol{\theta}|\mathcal{D}_{1:t-1})$, that is, we can utilize a recursive variational inference method to solve this problem. For example, the first posterior is formed as $p(\boldsymbol{\theta}|\mathcal{D}_1) = \text{proj}(p_0(\boldsymbol{\theta}) p(\mathcal{D}_1|\boldsymbol{\theta}))$, which can be approximated by the approximate posterior probability distribution $q_1(\boldsymbol{\theta})$. $q(\boldsymbol{\theta}) = \text{proj}(p(\boldsymbol{\theta}))$ denotes a projection operation that takes the intractable un-normalized distribution $p(\boldsymbol{\theta})$ and returns a tractable normalized approximation $q(\boldsymbol{\theta})$. After approximating the initial posterior distribution, subsequent estimations can be iteratively generated by amalgamating the approximate posterior distribution with the likelihood and projecting, that is, $p(\boldsymbol{\theta}|\mathcal{D}_{1:t}) = proj(q_{t-1}^*(\boldsymbol{\theta}) p(\mathcal{D}_t|\boldsymbol{\theta}))$, where $q_{t-1}^*(\boldsymbol{\theta})$ is the optimal approximate posterior distribution for Task $t-1$.

Here, we use Kullback–Leibler (KL) divergence to measure the gap between the true posterior and the approximate posterior $q_t(\boldsymbol{\theta})$. Minimizing the KL divergence that makes the approximate posterior $q_t(\boldsymbol{\theta})$ as close as possible to the true posterior $p(\boldsymbol{\theta}|\mathcal{D}_{1:t})$, ensuring that the model captures the underlying data distribution effectively across sequential tasks. we can get the optimal approximate posterior $q_t^*(\boldsymbol{\theta})$ for Task t

$$q_t^*(\boldsymbol{\theta}) = \arg\min_{q \in \mathcal{Q}} \text{KL}\left(q(\boldsymbol{\theta}) \| \frac{1}{Z_t} q_{t-1}(\boldsymbol{\theta}) p(\mathcal{D}_t \mid \boldsymbol{\theta})\right) \tag{5}$$

where $\mathcal{Q}$ is the set of allowed approximate posteriors. Z_t is the normalizing constant of $p_t^*(\boldsymbol{\theta}) = q_{t-1}(\boldsymbol{\theta}) p(\mathcal{D}_t|\boldsymbol{\theta})$, which can be ignored in the optimization process.

To make the computation and model's iterative process feasible, assuming the parameter θ is independent and identically distributed, we approximate the approximate posterior using a cluster of normal distributions, known as Gaussian mean field approximation [44]. Then, the approximate posterior distribution of parameters for the tth task $q_t(\boldsymbol{\theta})$ can be written as follows:

$$q_t(\boldsymbol{\theta}) = \prod_{d=1}^{N_t} \mathcal{N}\left(\boldsymbol{\theta}_{t,d}; \mu_{t,d}, \sigma_{t,d}^2\right). \tag{6}$$

3) Coreset Storage: During iterative approximation, as the number of tasks increases, errors from the approximate posterior distribution may accumulate significantly. To address this problem, we introduce the method of setting coreset. The coreset is analogous to an episodic memory that retains key information from the SAR images of previous tasks. The algorithm can reselect the coreset to update its memory of past tasks. For Task t, we choose SAR images from the collection of the current task dataset $\mathcal{D}_t$ and the old coreset C_{t-1} to form the new coreset C_t as shown in Fig. 2. We utilize the greedy K-center method [45] to select K unbiased coreset data C_t from $\mathcal{D}_t \cup C_{t-1}$ and guarantee that these data points are scattered over the input domain (data space over all tasks).

Through the introduction of coreset, analogizing to (3), the posterior probabilities we need can be decomposed into the equations related to C_t as follows:

$$\begin{aligned} q_t(\boldsymbol{\theta}) \to p(\boldsymbol{\theta} \mid \mathcal{D}_{1:t}) &\propto p(\boldsymbol{\theta} \mid \mathcal{D}_{1:t} \backslash C_t) p(C_t \mid \boldsymbol{\theta}) \\ &\approx \tilde{q}_t(\boldsymbol{\theta}) p(C_t \mid \boldsymbol{\theta}) \end{aligned} \tag{7}$$

where $p(\boldsymbol{\theta} \mid \mathcal{D}_{1:t} \backslash C_t) = \tilde{q}_t(\boldsymbol{\theta})$ in (7) refers to the posterior derived from data $\mathcal{D}_{1:t}$ excluding the new coreset C_t, while $p(C_t \mid \boldsymbol{\theta})$ means the likelihood of data in new coreset C_t.

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According to (3), the recursion of the posterior $\tilde{q}_t(\boldsymbol{\theta})$ can be rewritten as

$$\begin{aligned}\tilde{q}_t(\boldsymbol{\theta}) &\rightarrow p(\boldsymbol{\theta} \mid \mathcal{D}_{1:t} \backslash C_t) \\ &\propto p(\boldsymbol{\theta} \mid \mathcal{D}_{1:t-1} \backslash C_{t-1}) p(\mathcal{D}_t \cup C_{t-1} \backslash C_t \mid \boldsymbol{\theta}) \\ &\approx \tilde{q}_{t-1}(\boldsymbol{\theta}) p(\mathcal{D}_t \cup C_{t-1} \backslash C_t \mid \boldsymbol{\theta}) \\ &= p(\boldsymbol{\theta} \mid \mathcal{D}_{1:t-1} \backslash C_{t-1}) p(C_{t-1} \backslash C_t \mid \boldsymbol{\theta}) p(\mathcal{D}_t \backslash C_t \mid \boldsymbol{\theta}) \quad (8)\end{aligned}$$

where the previous posterior from data $\mathcal{D}_{1:t-1}$ excluding the prior coreset C_{t-1} is represented as $p(\boldsymbol{\theta}|\mathcal{D}_{1:t-1} \backslash C_{t-1})$. $p(C_{t-1} \backslash C_t|\boldsymbol{\theta})$ refers to the likelihood computed from data C_{t-1} outside the new coreset C_t. $p(\mathcal{D}_t \backslash C_t|\boldsymbol{\theta})$ indicates the likelihood derived from new data $\mathcal{D}_t$ not included in the newly updated coreset C_t. Hence, according to (8), we perform propagation via $\tilde{q}_t(\boldsymbol{\theta}) = proj(\tilde{q}_{t-1}(\boldsymbol{\theta}) p(\mathcal{D}_t \cup C_{t-1} \backslash C_t|\boldsymbol{\theta}))$ and employ the variational KL divergence to achieve the recursive variational inference

$$\begin{aligned}&\tilde{q}_t(\boldsymbol{\theta}) \\ &= \arg \min_{q(\boldsymbol{\theta}) \in Q} KL\left(q(\boldsymbol{\theta}) \| \frac{1}{Z_t} \tilde{q}_{t-1}(\boldsymbol{\theta}) p(\mathcal{D}_t \cup C_{t-1} \backslash C_t|\boldsymbol{\theta})\right). \quad (9)\end{aligned}$$

Through (7), we use the KL divergence to obtain the optimal approximate posterior $q_t^*(\boldsymbol{\theta})$ for ACRM's parameter estimation as follows:

$$q_t(\boldsymbol{\theta}) = \arg \min_{q(\boldsymbol{\theta}) \in Q} KL\left(q(\boldsymbol{\theta}) \| \frac{1}{Z_t} \tilde{q}_t(\boldsymbol{\theta}) p(C_t|\boldsymbol{\theta})\right). \quad (10)$$

By employing the above method, we directly integrate the coreset into the approximate posterior to obtain the prediction, thereby aiding in mitigating forgetting of the knowledge on the previous tasks. In a more comprehensive view, our approach is equal to the implementation of variational inference using information passing, wherein the coreset data points C_t are updated after the other tasks' data $\mathcal{D}_{1:t}$.

4) Overall Loss Function: In (10), we employ Monte Carlo sampling and reparameterization trick to compute the stochastic gradients [42]. According to [46], it can be derived that minimizing the KL distance is equivalent to maximizing the following loss function:

$$\begin{aligned}&\mathcal{L}_{\mathrm{ACRM}}^t(q_t(\boldsymbol{\theta})) \\ &= \sum_{n=1}^{N_t} \left[-y_t^{(n)} \log\left(\hat{y}_t^{(n)}\right) + \left(1 - y_t^{(n)}\right) \log\left(1 - \hat{y}_t^{(n)}\right)\right] \\ &\quad - \mathrm{KL}(q_t(\boldsymbol{\theta}) \| q_{t-1}(\boldsymbol{\theta})) \\ &= \sum_{n=1}^{N_t} \mathbb{E}_{\boldsymbol{\theta} \sim q_t(\boldsymbol{\theta})}\left[\log p\left(y_t^{(n)} \mid \boldsymbol{\theta}, \mathbf{x}_t^{(n)}\right)\right] \\ &\quad - \mathrm{KL}(q_t(\boldsymbol{\theta}) \| q_{t-1}(\boldsymbol{\theta})). \quad (11)\end{aligned}$$

The first term in (11) is the cross-entropy loss in target recognition, which measures the accuracy. The second term is the regularization constraint in online variational inference, which restricts the variation of the model's parameters and improves the generalization of the model.

The propagation of ACRM-Coreset is achieved by the recurrence of $\tilde{q}_t(\boldsymbol{\theta})$, which is continually updated through (9). The prediction is obtained by the distribution $q_t(\boldsymbol{\theta})$ through (10). During training, $q_t(\boldsymbol{\theta})$ and C_t are combined to get the final prediction distribution. For the test, SAR images are directly fed into the model without the setting of the coreset. The overall calculation process is described in Algorithm 1.

Algorithm 1 Overall Calculation Algorithm

Input: Prior distribution $p(\boldsymbol{\theta})$, coreset size K from each task

Output: Approximate posterior and prediction distribution for each task $\{q_t(\boldsymbol{\theta}), p(y^*|\boldsymbol{x}^*, \mathcal{D}_{1:t})\}_{t=1}^T$

Initialize the coreset and approximate posterior: $C_0 \leftarrow \varnothing, q_0 \leftarrow q_0^*$;

for $t = 1, 2, \ldots T$ **do**

  Observe the dataset $\mathcal{D}_t$ for next task ;

  Utilize C_{t-1} and $\mathcal{D}_t$ to update the coreset C_t ;

  Update the approximate posterior $\tilde{q}_t(\boldsymbol{\theta})$ for data points $\mathcal{D}_t \cup C_{t-1} \backslash C_t$ by (9) (used for propagation);

  Derive the final approximate posterior through 10 (only utilized in prediction, but not in propagation); Get category prediction for test SAR images $\boldsymbol{x}$: $p(\hat{y} \mid \boldsymbol{x}, \mathcal{D}_{1:t}) = \int q_t(\boldsymbol{\theta}) p(\hat{y} \mid \boldsymbol{\theta}, \boldsymbol{x}) \mathrm{d}\boldsymbol{\theta}$;

  Update t;

end

IV. Experiments

This section introduces experimental settings and results. We demonstrate ACRM's ability to retain information on old tasks and learn new ones on a toy dataset, as well as on the MSTAR and FUSARship datasets separately. We demonstrate the stability of ACRM by varying the coreset size. We make a comparison with several baseline continual learning methods to prove the ACRM's effectiveness and robustness.

A. Dataset Description

We use the MSTAR and FUSARship datasets in our experiments.

1) *MSTAR*: This dataset is provided by Sandia National Laboratory, including SAR images captured with X-band radar worked in HH polarization. It contains a variety of different military targets, such as the T72 tank, BTR-70 armored vehicles, and others.
2) *FUSARship*: This dataset comprises 15 main ship classes and 98 subclasses, along with various nonship marine targets. It derives from 126 Gaofen-3 remote sensing images. These images possess DH and DV polarization modes and have a resolution of 1.124 × 1.728 m in ultrafine resolution SAR imaging mode. It covers diverse environments including the sea, land, coastal areas, rivers, and islands.

In both datasets, training (Tr) and testing (Te) SAR images are divided into different target-aspect domains: [0°, 90°), [90°, 180°), [180°, 270°), and [270°, 360°), enabling a robust evaluation of continual learning algorithms in varying conditions. Both datasets are augmented using the synthetic minority over-sampling technique (SMOTE) [48]. Table I displays the

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TABLE I

SAMPLE PRESENTATION OF THE MSTAR DATASET AND THE FUSARSHIP DATASET

Class / Domains	MSTAR							FUSARship				
	2S1	BRDM2	BTR60	D7	T62	ZIL131	ZSU234	BulkCarrier	ContanierShip	Fishing	GeneralCargo	Tanker
	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te	STS Tr Te
[0°, 90°)	300 71	300 72	300 68	300 70	300 72	300 72	300 72	300 24	300 20	300 37	300 57	300 54
[90°, 180°)	300 74	300 73	300 68	300 82	300 74	300 74	300 73	300 22	300 18	300 19	300 27	300 31
[190°, 270°)	300 79	300 80	300 60	300 68	300 80	300 80	300 80	300 20	300 20	300 34	300 41	300 32
[270°, 360°)	300 75	300 73	300 60	300 79	300 72	300 73	300 74	300 16	300 17	300 15	300 34	300 33

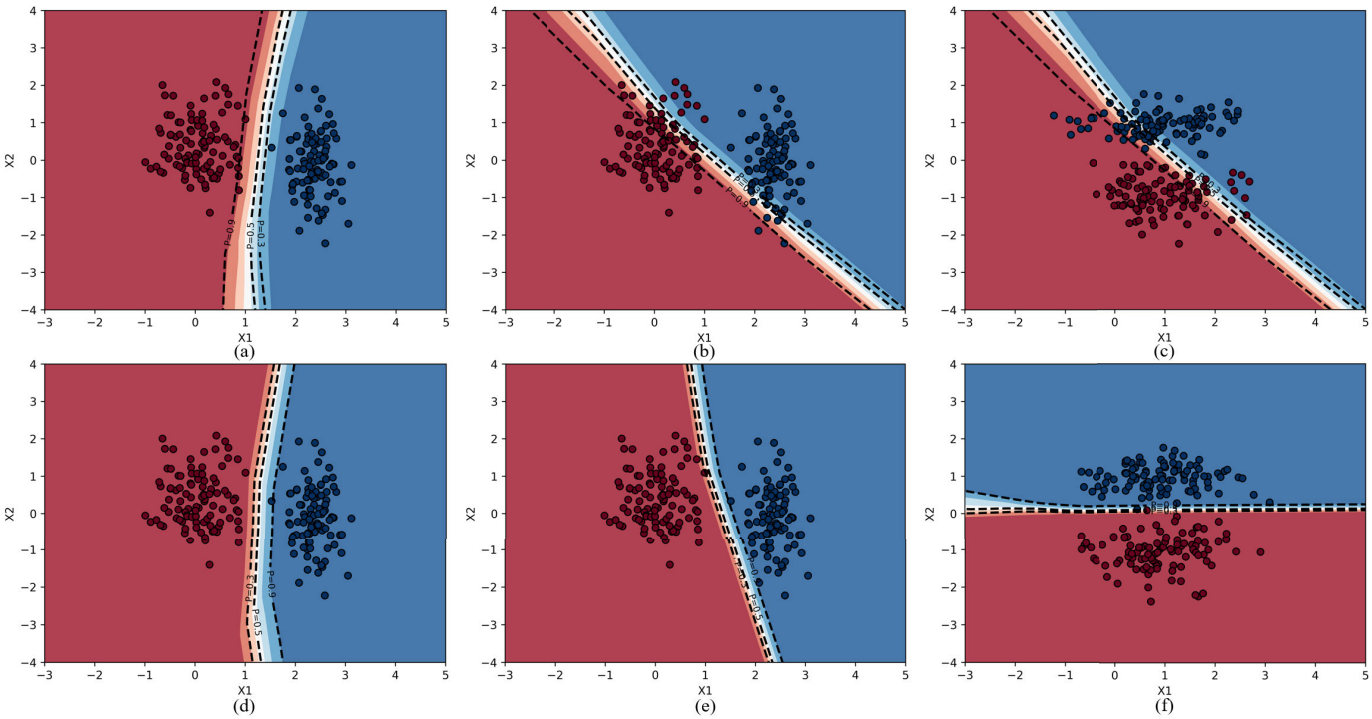


Fig. 5. Online learning ability of ACRM and the recognition model with EWC trained on 2-D toy datasets. (a) and (d) Show the recognition models' classification boundaries for Task 1 in the training order: Task 1. (b) and (e) Show the recognition models' classification boundaries for Task 1 in the training order: Task 1 and Task 2. (c) and (f) Show the learned recognition models' classification boundaries for Task 2 in the training order: Task 1 and Task 2. (a)–(c): recognition model with EWC. (d)–(f): ACRM.

SAR target samples (STSs) across four target-aspect domains, detailing the sizes of the Tr and Te datasets.

B. Experimental Settings

In each training phase, the model is trained for 100 epochs with a learning rate of 0.001. In the experiment, the SAR image sizes are all set to 100 × 100, and the mini-batch size is set to 32. The basic deep network is ResNet-18. We take ReLU as the activation function between each layer and use the activation function Softmax in the output layer. We optimize all models using the Adam optimizer. All experiments are implemented in Python using PyTorch and executed on a computer with an Intel Core i9-13900HX CPU and NVIDIA RTX 4090 GPU.

In the experiment, mean accuracy stands for the mean test accuracy of Monte Carlo repeated experiments, and average accuracy means the recognition model's average test accuracy on Tasks 1 to t, which are online trained on the training SAR images for Task t. We compare our algorithms with some baseline continual learning algorithms.

1) *Elastic Weight Consolidation*: Kirkpatrick et al. [35] adopted a Fisher information matrix to calculate the importance of network parameters.
2) *Learned Context-Dependent Gating (LXDG)*: Tilley et al. [37] introduced a method that can flexibly allocate active neurons in the hidden layer combined with a new regularization term.
3) *Learning Without Forgetting (LWF)*: Li and Hoiem [40] used the predictions from the old tasks to compute the distillation loss while learning new samples.
4) *Learning Without Prejudice*: Jeon et al. [41] raised a learning method to encourage benign forgetting which is favorable by the model and suppresses harmful forgetting.

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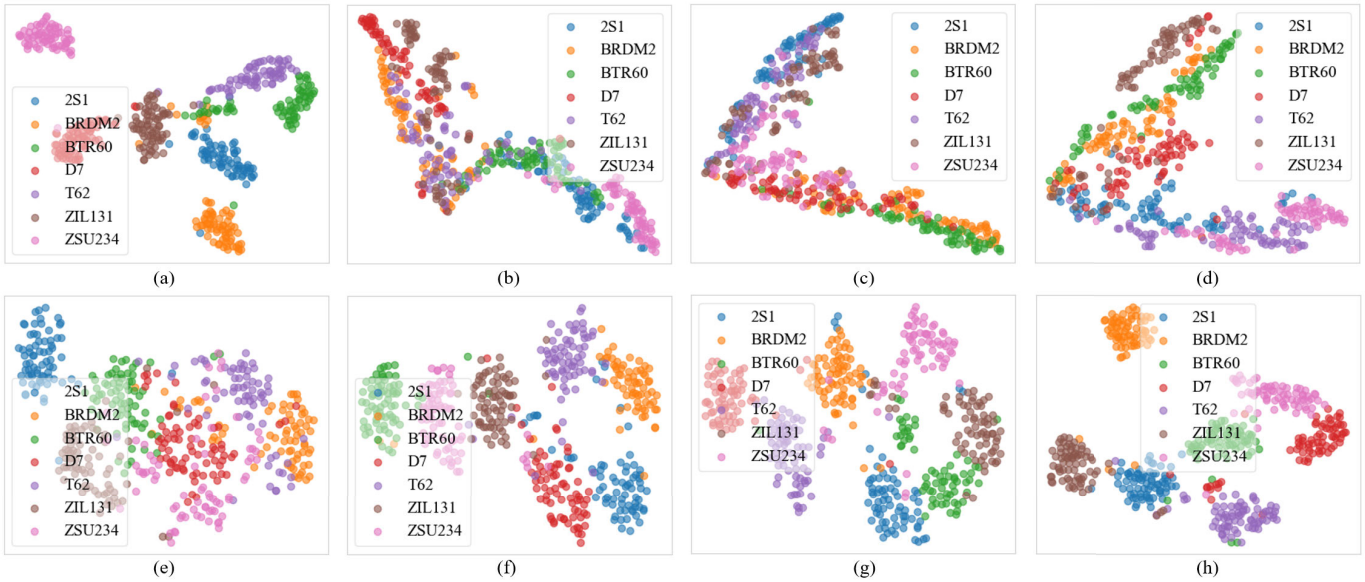


Fig. 6. Using t-SNE to reduce the feature dimensions to 2, we compare the separability of features extracted by ACRM and the recognition model with EWC, based on training SAR images from Tasks 1–4 of the MSTAR dataset. (a)–(d) Recognition model with EWC. (e)–(h) ACRM.

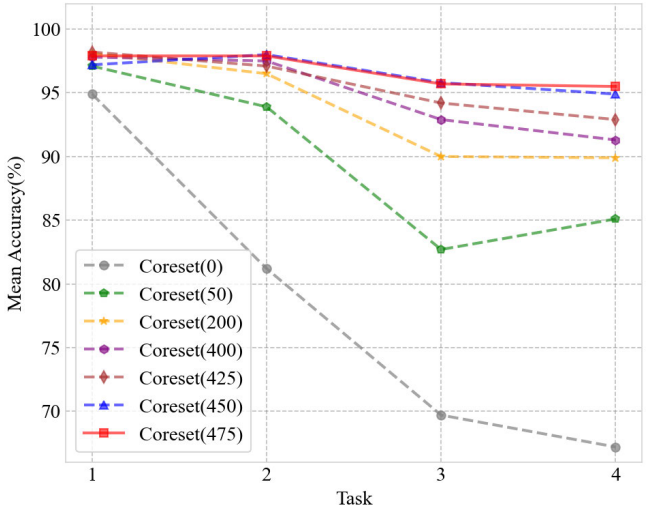


Fig. 7. Average accuracies of ACRM-Coreset for various coreset sizes on the MSTAR dataset.

5) *Hard Attention*: Serra et al. [39] used a node-based hard attention (HAT) mechanism, which utilizes an attention-weighted regularization through gating task embedding.
6) *Context-Dependent Gating (XDG)*: Masse et al. [36] proposed the use of logic gate control to maintain high performance across large numbers of sequentially presented tasks.
7) *Synaptic Intelligence (SI)*: Zenke et al. [38] put forward the model that prevents old memories from being overwritten by penalizing changes of important parameters.

C. Experiment on the Toy Dataset

We experiment on a 2-D toy dataset to compare the performance of the recognition model with elastic weight consolidation (EWC) and ACRM. We test them on two sequential classification tasks. The two tasks generate two types of Gaussian distribution points as the data samples of two classes. There are 200 data samples per task and 100 data samples per class. The data samples of the first class obey the same Gaussian distribution for both tasks, which are marked with red. However, the data samples of the second class obey the different Gaussian distributions for both tasks, which are marked with blue.

For comparison, the ACRM and the recognition model with the EWC are both continually trained on the toy dataset from Task 1 to Task 2. We use a simple fully connected network with a single hidden layer of 20 units and ReLU activation for both models.

Fig. 5(a) and (d) shows the recognition model's classification boundaries (the prediction probability contours of the first class), which are trained on the training data of Task 1. The two models successfully classify for Task 1, when trained on the training samples of Task 1. Fig. 5(b) and (c) shows the recognition model with EWC's classification boundaries for Task 1 and Task 2, respectively, when trained on the training samples of two tasks. Fig. 5(e) and (f) shows the recognition model's classification boundaries by ACRM. It can be observed that when the second task comes, the recognition model with EWC is unable to distinguish the two classes well for both tasks, while ACRM can predict the classification boundaries excellently. Since our algorithm has the merit of both neural networks and Bayesian theory, which enhances the generalization capability of the recognition model for continual learning.

D. Separability of the Extracted Features

To further demonstrate the prominent feature separability of ACRM. Fig. 6 compares the separability of features extracted by ACRM and the recognition model with EWC, using t-SNE [47] to reduce the features to 2 dimensions. Fig. 6(a)–(d) and (e)–(h) shows the change of

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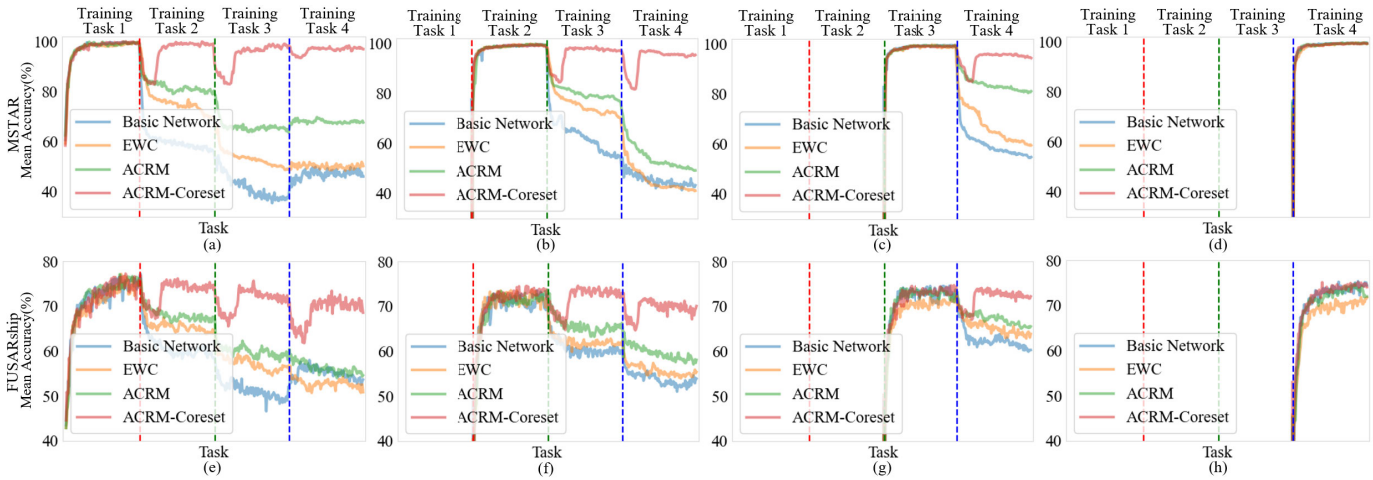


Fig. 8. Fluctuation of different models' test accuracies on Task t (t = 1:4), which are online trained on Task t to 4. Results are depicted for the MSTAR dataset (a)–(d) and the FUSARship dataset (e)–(h).

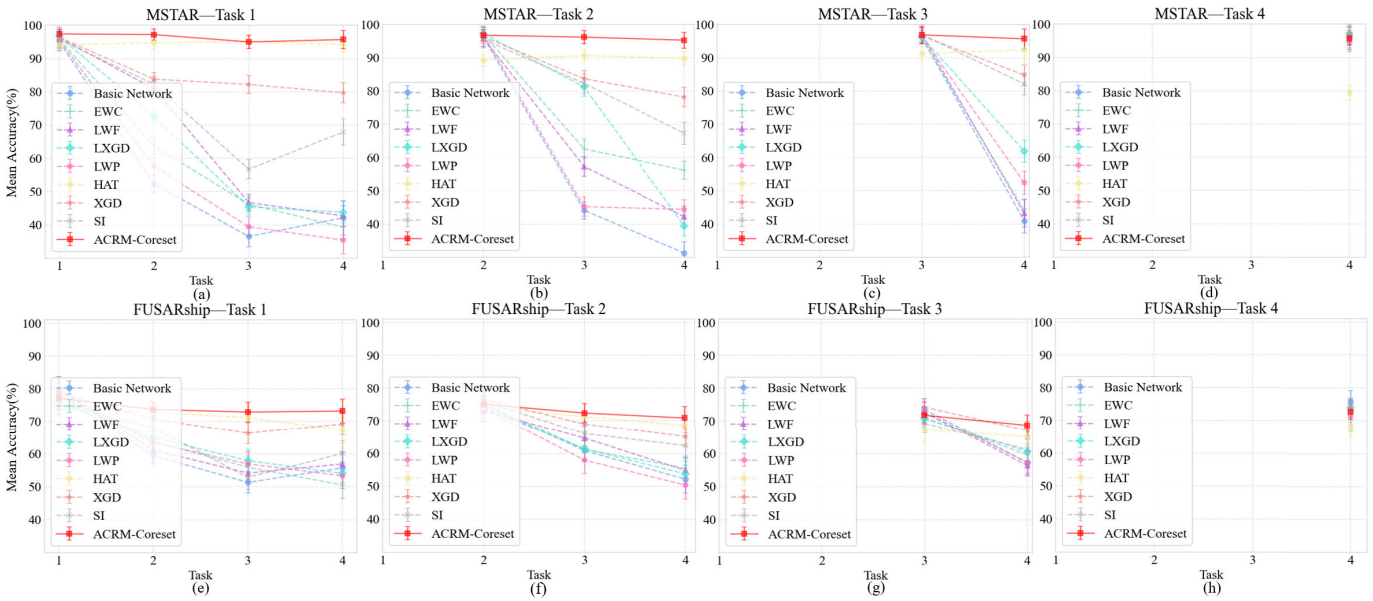


Fig. 9. Different models' mean test recognition accuracies for Task t (t = 1:4), which are online trained on the training SAR images of Task t to 4. Error bars at data points are 95% confidence intervals. Results are depicted for the MSTAR dataset (a)–(d) and the FUSARship dataset (e)–(h).

feature separability for Task 1 after the recognition model with EWC and ACRM observing the samples from Tasks 1 to 4, respectively.

The separability of the features learned by ACRM decreases slightly as the number of arriving tasks increases, while the separability of the features learned by the recognition model with EWC decreases more significantly. ACRM models the SAR images through probability distribution, which can describe the varied SAR images more sufficiently and reduce forgetting the knowledge learned from the previous tasks.

E. Impact of the Coreset Size on Model Performance

To investigate the effect of coreset size on ACRM performance, we vary the sizes of coresets in the range of 0–475. As shown in Fig. 7, recognition accuracy remains high on the MSTAR dataset for coreset sizes between 450 and 475. By calculating the average accuracy, we can assert that the overall accuracy maintains a commendable level when the coreset size is large. Considering the tradeoff between calculating speed and recognition accuracy, the following experiments are conducted with a coreset size of 475.

F. Recognition Accuracy and PDR

In Fig. 8, we present the recognition accuracies for four distinct methods applied to the MSTAR and FUSARship datasets across multiple training tasks. It can be observed that, in terms of recognition precision, the recognition model with the EWC outperforms the basic recognition model. This superiority is due to EWC's regularization term, which restricts parameter updates and helps retain knowledge from previous tasks. Our noncoreset approach surpasses the model with EWC in recognition performance, as it employs Bayesian theory to model the varied distribution of scattering points in SAR images, thereby enhancing the model's generalization for continual target recognition.

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TABLE II
MEAN ACCURACIES VERSUS PDR FOR VARIOUS METHODS ON THE MSTAR DATASET ACROSS MULTIPLE TASKS

Method	T=1(%)	T=2(%)		T=3(%)			T=4(%)				Avg.acc	PDR ↓
	Task1	Task1	Task2	Task1	Task2	Task3	Task1	Task2	Task3	Task4		
Basic-Network	**97.1**	52.3	96.2	36.5	44.1	96.2	42.1	31.3	40.9	96.3	52.6	55.0
EWC	95.8	63.7	97.1	46.2	62.6	95.9	39.3	56.2	44.1	**97.2**	59.2	56.5
LWF	96.3	80.9	95.9	46.7	57.3	**97.1**	42.7	42.2	43.2	95.3	55.8	53.6
LXGD	96.6	72.6	96.8	45.4	81.2	96.3	43.8	39.5	61.9	97.1	60.5	52.8
LWP	95.5	57.6	97.0	39.4	45.2	96.5	35.4	44.4	52.4	95.9	57.0	60.1
HAT	94.2	94.7	89.3	**95.1**	90.7	91.3	93.9	89.8	92.4	80.2	89.1	**0.3**
XGD	96.4	83.8	96.3	82.2	83.7	96.6	79.7	78.2	84.8	96.9	84.9	16.7
SI	95.7	82.3	95.5	56.7	82.3	97.0	67.9	67.3	82.1	94.5	77.9	27.8
ACRM	95.8	68.3	95.3	45.1	72.3	95.2	55.3	46.8	73.4	96.4	67.9	40.5
ACRM-Coreset	97.0	**97.1**	**97.3**	95.0	**96.2**	96.9	**95.7**	**95.3**	**95.7**	95.4	**95.5**	1.3

TABLE III
MEAN ACCURACIES VERSUS PDR FOR VARIOUS METHODS ON THE FUSARSHIP DATASET ACROSS MULTIPLE TASKS

Method	T=1(%)	T=2(%)		T=3(%)			T=4(%)				Avg.acc	PDR ↓
	Task1	Task1	Task2	Task1	Task2	Task3	Task1	Task2	Task3	Task4		
Basic-Network	78.4	59.4	74.8	51.3	60.8	73.6	55.7	52.1	57.2	**75.7**	60.1	22.7
EWC	79.2	63.7	75.3	55.9	61.2	71.1	50.5	55.4	59.6	69.6	58.7	28.7
LWF	**79.9**	61.1	73.1	54.1	64.6	72.7	56.9	54.8	56.3	73.8	60.4	23.0
LXGD	75.8	65.0	73.9	57.9	61.3	70.8	54.0	53.7	60.6	74.0	60.5	21.8
LWP	78.1	63.4	73.7	56.9	57.9	72.7	53.3	50.3	57.3	71.6	58.1	24.8
HAT	75.2	72.7	74.0	71.2	71.3	67.3	67.3	68.6	65.2	67.9	67.2	7.9
XGD	78.9	70.7	**76.6**	66.3	69.0	**74.3**	69.2	65.1	57.1	74.0	66.3	9.7
SI	75.4	68.0	76.0	52.8	66.0	69.3	60.1	62.3	61.2	73.3	64.2	15.3
ACRM	76.5	64.8	74.2	58.7	65.1	71.3	54.9	59.9	66.4	71.8	63.2	21.6
ACRM-Coreset	77.1	**73.7**	75.1	**72.9**	**72.4**	71.8	**73.2**	**70.9**	**68.7**	72.7	**71.3**	**3.9**

During the initial phase of training, there is a noticeable performance decline for all tasks. This decline occurs because, at this stage, the SAR images are used to train the ACRM without the coreset C_t. Subsequently, the recognition accuracies for different tasks increase significantly when the coreset C_t is incorporated into the training dataset. The inclusion of the coreset aids in recalling SAR images from previous tasks, thereby mitigating the forgetting of learned knowledge. Moreover, the ACRM-Coreset surpasses the ACRM because the coreset component reduces the buildup of estimation errors that arise from the iterative approximations during inference in ACRM.

In Fig. 9(a)–(d) and (e)–(h), we compare the mean test recognition accuracies of different algorithms for the current Task t ($t = 1{:}4$). These algorithms are trained online using the SAR images from training Task t to Task 4. Tables II and III present the mean recognition accuracies of various algorithms over each task individually. Specifically, for the four columns in part $T = 4$, the models are trained sequentially on the SAR images of Tasks $T = 1$ to 4. After training on Task $T = 4$, the recognition models are tested on the SAR images of Tasks 1–4, respectively.

We observe that, after training on the current task, the accuracies for previous tasks decline for all algorithms. This trend is consistent with that in Figs. 8 and 9 and Tables II and III. Most algorithms fail to maintain recognition performance across all tasks, except for HAT, XGD, and ACRM-Coreset. HAT utilizes attention vectors from previous tasks to create a binary mask, keeping some weights static, while others adapt to new tasks. XGD uses a context-dependent gating signal combined with weight regularization to mitigate forgetting. The basic network, which lacks any continual learning mechanism, exhibits the most significant decrease in recognition precision. Learning without prejudice (LWP) experiences the second-largest decline because its constructed generator does not effectively handle SAR images, confusing the recognition model. In contrast, ACRM-Coreset is the most stable and maintains excellent recognition performance by recalling learned knowledge from previous tasks through the selection of important samples as the coreset.

The performance drop rate (PDR), as described by Zhang et al. [49], serves as an effective measurement for assessing how well an algorithm mitigates the forgetting of previously learned knowledge in continual SAR recognition tasks. The PDR quantifies the reduction in recognition accuracy, calculated as follows:

$$\text{PDR} = A_1 - A_T. \tag{12}$$

Here, A_1 represents the recognition accuracy achieved when the model is trained solely on Task 1 and A_T denotes the accuracy on Task 1 after the model has been trained up to Task t.

Tables II and III present the PDRs and average test accuracies of the algorithm on Task 1, trained online on SAR images from Tasks 1 to 4. Notably, HAT exhibits the lowest PDR on the MSTAR dataset, whereas ACRM-Coreset achieves the lowest PDR on the FUSARship dataset. Additionally, ACRM-Coreset not only maintains the low PDRs but also sustains the highest average recognition accuracies across the tasks.

G. Impact of Task Order on Model Performance

To investigate the influence of training task orders on the ACRM's performance in SAR-ATR across various target-aspect domains, we use three SAR images tasks in

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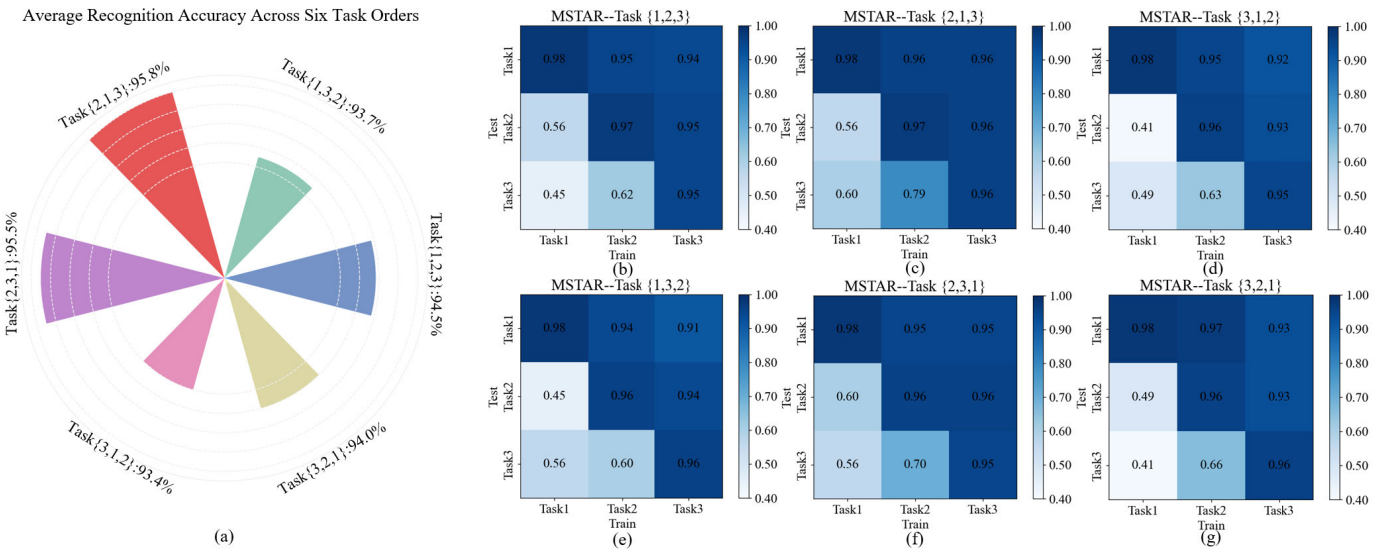


Fig. 10. Performance of ACRM across six task orders on the MSTAR dataset. (a) Average accuracy of ACRM on the comprehensive test set following the completion of training for different task orders. (b)–(g) Mean accuracy ACRM on every task test set for each specified training order.

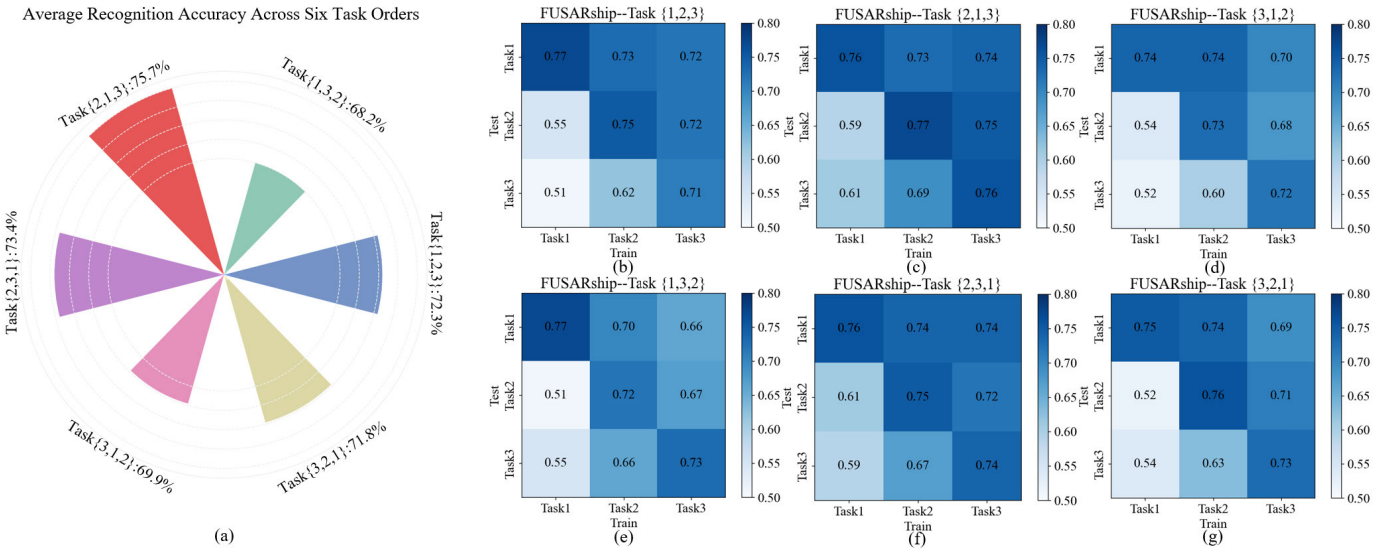


Fig. 11. Performance of ACRM across six task orders on the FUSARship dataset. (a) Average accuracy of ACRM on the comprehensive test set following the completion of training for different task orders. (b)–(g) Mean accuracy ACRM on every task test set for each specified training order.

Table I with domains of $[0°, 270°)$ to constitute six task orders: Task{1,2,3}, Task{1,3,2}, Task{2,1,3}, Task{2,3,1}, Task{3,1,2}, and Task{3,2,1}. We show the results on the MSTAR and FUSARship datasets in Figs. 10 and 11.

Interestingly, when we first train ACRM on the SAR images of Task 2, followed by training on the SAR images from the remaining two tasks, the average recognition performance of the model outperforms that of the model trained according to the other four training task orders. This is because the sum of the differences in the distribution of the SAR target scatterers between Task 2 and the remaining 2 tasks is smaller than the sum of the differences in the distribution of the scatterers between Task 1 or 3 and the remaining tasks. By training Task 2 first, the parameter conflicts of the model in learning the remaining tasks (Task 1 and Task 3) are reduced, and the features learned from Task 2 are conducive to the model transferring to the remaining tasks.

H. Performance of Continual Learning in Mitigating the Clever Hans Effect Induced by Clutter

To explore the effectiveness of continual learning in mitigating the Clever Hans effect (where models fail to learn the true target features and instead rely on irrelevant features or biases in the data, leading to predictions influenced by noise or spurious correlations), we generate training SAR images with different intensities of clutters for three tasks. This is achieved by multiplying the SAR images with simulated clutter, which is sampled from Gamma distributions with varying strength parameters. The parameters of the Gamma distributions for the three tasks are shown in Fig. 12. The model is trained on the SAR images of tasks in the following order: Task 1, Task 2, and Task 3 (from high-intensity clutter to weak-intensity clutter), and tested on those of Task 1.

As shown in Fig. 13, due to the influences of clutter, the recognition models cannot learn enough features of the target areas, which leads to the poor performance of all models in

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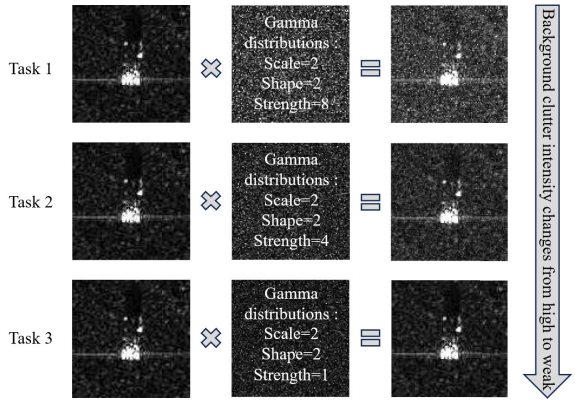


Fig. 12. Task division based on background clutter intensity range.

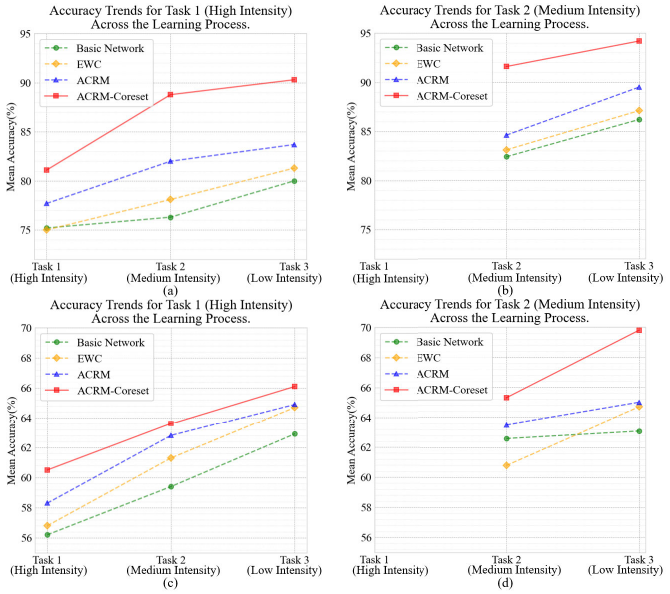


Fig. 13. Performance of ACRM online trained on SAR images with three coherent spot noise intensities shown in Fig. 12. Results for the MSTAR dataset are shown in (a) and (b), and for the FUSARship dataset in (c) and (d).

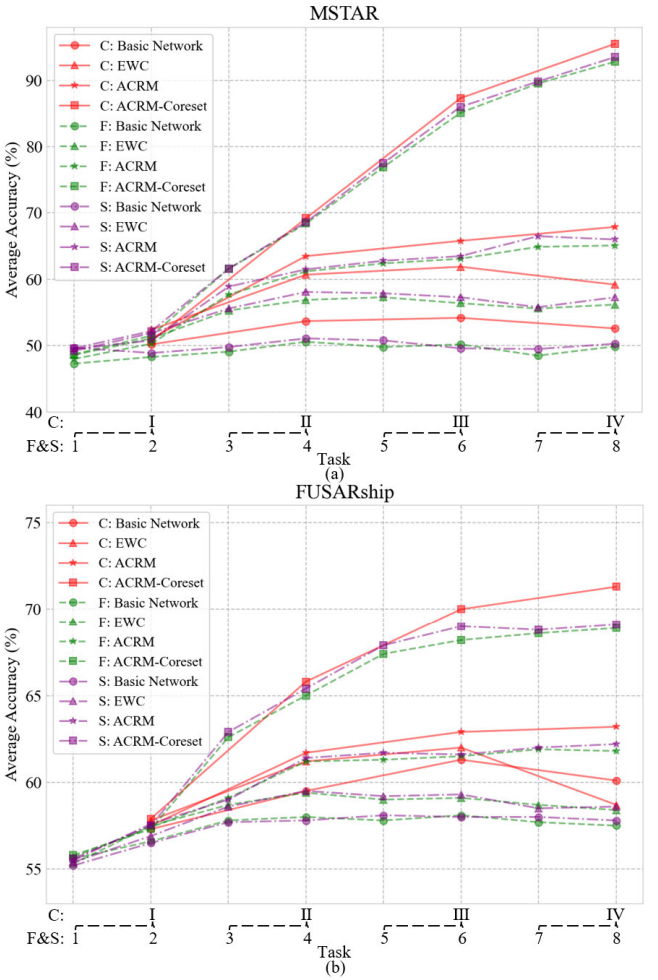


Fig. 14. Comparison of continual learning performance across coarse-grained and fine-grained tasks, where the target-aspect domain spans are divided by 90° and 45°, respectively. Group C consists of four coarse-grained tasks with a span of 90°, group F consists of eight fine-grained tasks with a span of 45°, and group S consists of eight fine-grained tasks with a span of 45° after speckle filtering. Results are depicted for the MSTAR dataset (a) and the FUSARship dataset (b).

Task 1. From the red line (ARCM-Coreset) and green line (Basic Network) in Fig. 13(a) and (c), through continuous learning SAR images with lower intensity clutter, the influence of clutter in SAR image recognition with high-intensity clutter ARCM-Coreset effectively is reduced, that is, the Clever Hans effect induced by clutter is mitigated. Compared to other methods, ACRM-Coreset has the best performance in all tasks. In detail, ACRM-Coreset performances better than ACRM, because coreset can mitigate the accumulation of estimation errors caused by iterative approximations in the inference process. The recognition model with EWC is worse because the ACRM uses a Bayesian framework to adequately fit the distributions of SAR targets' scattering points at different intensities of background clutter, unlike EWC which only constrains the parameter values of the model. The basic network has the worst performance due to lacking the ability to suppress catastrophic forgetting.

I. Impact of Fine-Grained Task Partitioning on Continuous Learning

To examine the effects of fine-grained task partitioning on ACRM performance, we divide all target-aspect domain data into four coarse-grained tasks (C) with a span of 90° and eight fine-grained tasks (F) with a span of 45°. Additionally, to investigate the impact of the speckle effect on ACRM's performance, we preprocess the fine-grained tasks into spotless tasks (S) using the SAR image speckle filtering with context covariance matrix formulation proposed by Chen [50].

As shown in Fig. 14, with the increase in the number of training tasks, the recognition models' performance in Group F becomes progressively worse compared to its performance in Group C. This is because the granulating of tasks leads to a decrease in the number of samples in each task and an increase in the number of tasks. The recognition model performs better in Group F than in Group S, as the speckle filter helps reduce the impact of clutter, thereby enhancing the overall recognition performance.

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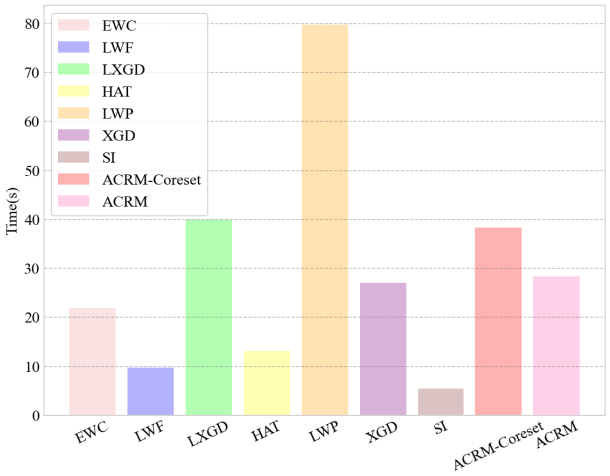


Fig. 15. Different algorithms' time consumption.

J. Time Consumption

In an actual battlefield environment, the computing time of the model can impact its capacity for monitoring. As shown in Fig. 15, we compare the time it takes on different methods to run an epoch with the same experimental settings. LWP costs the most time, due to its need to generate SAR images. However, SI costs the least time, because it utilizes a simple structure to calculate the amount of changes in the loss function's output values. With the addition of Monte Carlo sampling, ACRM costs a bit more time than the recognition model with EWC, but its recognition performance is significantly better. The addition of coreset in ACRM allows for a small increase in computing time but achieves better recognition performance furthermore. It can be summarized that ACRM-Coreset can make a desirable tradeoff between computing efficiency and recognition performance.

V. Conclusion

When SAR images with different target aspects are presented sequentially in the real-world SAR-ATR, to address the challenge of target-aspect feature forgetting, we propose ACRM based on an online variational Bayesian framework. In more detail, considering the abundant variations of SAR images with different target aspects, we utilize the Bayesian probabilistic frame to model the varied target features across different aspects and improve the model's generalization. To acquire a better solution for the posterior probability of the model parameters, we integrate an online Monte Carlo variational inference into the deep neural network in ACRM. Furthermore, we select the coreset by retaining a limited number of typical samples from previous tasks as a coreset and adding them to the training dataset for the current task, which can mitigate the accumulation of estimation errors caused by the repetitive approximations in inference and reduce the forgetting of knowledge learned from the previous tasks.

We conducted extensive experiments on the MSTAR and FUSARship datasets, comparing ACRM and ACRM-Coreset with various baseline continual learning methods, to demonstrate their superior continual recognition abilities. The results emphasize the capabilities of our model which can learn the features from the SAR images of the new target-aspect domain and avoid forgetting the learned features from the previous target-aspect domains.

Future research will focus on developing efficient training and inference methods to reduce computational resource consumption. We will also explore scenarios involving the continual arrival of both new target-aspect domains and categories.

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